Regime-Switching GARCH-LSTM. A Framework for Cryptocurrency Risk Management

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Abstract

This article presents RS-GARCH-LSTM, an extension of the GARCH-LSTM hybrid framework introduced by zhao2024 to the cryptocurrency volatility forecasting problem. The key innovation is the replacement of a constant scalar mixing weight with a dynamic gate w_t conditioned on six publicly available Bitcoin on-chain signals, allowing the model to adaptively allocate influence between the GARCH econometric baseline and the LSTM correction component on a daily basis. The model is evaluated across three experimental settings covering structurally different market conditions: BTC-USD in a normal out-of-sample period, BTC-USD in an unseen stress year (2022, comprising the LUNA and FTX collapses), and ETH-USD in a low-volatility period. RS-GARCH-LSTM is the only model that passes the Kupiec unconditional coverage test at 99% confidence on BTC in normal conditions ($p = 0.063$), achieves the highest Kupiec p-value on the unseen 2022 stress scenario ($p = 0.501$), and delivers a 34.8% reduction in capital efficiency ratio relative to GJR-GARCH on ETH in a period where calibration itself loses discriminating power. The gate weights, learned end-to-end without prior constraints on sign or magnitude, exhibit direct economic interpretability and are consistent with established on-chain analysis theory. On a representative \$100M BTC position under Basel III internal model frameworks, the correctly calibrated RS-GARCH-LSTM model releases approximately \$19.2M of excess capital relative to GJR-GARCH, corresponding to an annual opportunity cost of roughly \$960K at a 5% hurdle rate.

1 Motivation and Research Context

The Value-at-Risk (VaR) model underpinning a fund's daily capital allocation has direct balance-sheet consequences. Under Basel III Pillar 1 internal model frameworks, the market risk capital requirement is determined by the formula $\text{MRC}_t = k \cdot \max(\overline{\text{VaR}}_{60}, \widehat{\text{VaR}}_t)$ with supervisory multiplier $k \geq 3$. A model that systematically overestimates tail risk forces the institution to hold excess margin that cannot be deployed productively, at a cost that compounds daily over the full investment horizon.

The empirical failure of GJR-GARCH on cryptocurrency data is well-documented and quantifiable. On the BTC-USD test period (March 2025 to May 2026, $N = 415$ days), the model produces zero violations at the 99% level, yielding a Kupiec LR test p-value of 0.004 (rejected at 0.5%). The same failure mode recurs on the 2022 bear market scenario ($p = 0.007$). Zero violations imply a systematic over-estimation of tail risk: the VaR is permanently wider than the empirical loss distribution requires. On a \$100M BTC position, this inflates the Basel III capital requirement from \$33.3M (correctly calibrated) to \$55.6M, locking \$19.2M in unproductive margin for no measurable reduction in actual tail exposure.

Three structural properties of cryptocurrency markets explain this failure. First, Bitcoin and Ethereum exhibit near-integrated volatility persistence ($\hat{\beta} \approx 0.89$–0.91) and extremely leptokurtic return distributions ($\kappa > 14$), which cause maximum-likelihood GARCH estimates to be persistently conservative. Second, regime transitions in crypto markets are frequently preceded by observable on-chain signals, including miner capitulation events (falling hash rate), mempool congestion (rising transaction fees), and unusual capital inflows (transaction volume spikes), all of which manifest in blockchain data before they are visible in price returns. Third, GARCH model parameters are constant by construction: the same $(\hat{\omega}, \hat{\alpha}, \hat{\lambda}, \hat{\beta})$ that fit a low-volatility expansion period are applied unchanged to a structural market collapse without any mechanism for recalibration.

This article introduces RS-GARCH-LSTM as a minimal and interpretable solution to these three limitations simultaneously. The remainder is organised as follows. Section 2 describes the theoretical foundation from zhao2024. Section 3 presents the RS-GARCH-LSTM architecture and training methodology. Section 4 describes the data and experimental setup. Sections 5 and 6 report the full empirical results for BTC-USD and ETH-USD respectively. Section 7 translates the statistical results into financial and risk-management implications. Section 8 concludes.

2 Theoretical Foundation: The GARCH-Neural Network Equivalence

zhao2024 established that a GARCH(1,1) model is mathematically equivalent to a recurrent neural network with a scalar input, no output layer, and no tanh activation. This equivalence is not merely of theoretical interest. It implies that the GARCH family can be embedded within a broader neural architecture that preserves the econometric structure and interpretability of GARCH while adding the non-linear correction capacity of a long short-term memory (LSTM) network.

The Zhao et al. model is:

$$\sigma_t^2 = \underbrace{\mathcal{K}_{\text{GJR}}(\varepsilon_{t-1}, \sigma_{t-1}^2;\, \Theta)}_{\text{GJR-GARCH baseline}} \cdot \big(1 + w \cdot \tanh(c_t)\big), \tag{1}$$

where $\mathcal{K}_{\text{GJR}}$ is the GJR-GARCH(1,1) kernel glosten1993, $\Theta = (\omega, \alpha, \lambda, \beta)$ are the GARCH parameters, c_t is the LSTM cell state updated from past returns and variances, and $w \geq 0$ is a constant scalar mixing weight. When $w = 0$ the model reduces exactly to GJR-GARCH. When $w > 0$ the LSTM adds a multiplicative, time-varying correction to the GARCH baseline. Zhao et al. demonstrate that this hybrid outperforms both pure GARCH specifications and Transformer-based architectures on five equity and foreign exchange benchmarks (S&P 500, DJIA, NASDAQ, EUR/USD, Gold) on mean absolute error (MAE) and mean squared error (MSE).

Applying this framework to BTC/USD (2017–2026) with six on-chain exogenous inputs extends the LSTM input from $[\varepsilon_{t-1}, \sigma^2_{t-1}]$ to $[\varepsilon_{t-1}, \sigma^2_{t-1}, X_{t-1}]$. The out-of-sample performance improvement over GJR-GARCH is substantial: MAE -28.1%, RMSE -17.6%, $\Delta R^2 = +0.33$. The Optuna-optimal weight converges to $w = 0.084$. Crucially, the GARCH parameters learned end-to-end ($\hat{\beta}_{\text{e2e}} = 0.892$) reproduce the standalone MLE estimates ($\hat{\beta}_{\text{MLE}} = 0.894$) to three decimal places, providing empirical validation of the Zhao equivalence on cryptocurrency data.

The structural limitation of this extended model is the constant weight w. During the LUNA collapse in May 2022, blockchain data exhibited clear indicators of structural stress: hash rate declining under miner financial pressure, total transaction fees spiking on mempool congestion, and transaction volume surging as market participants repositioned. All of this information entered the LSTM through the exogenous input channel X_{t-1}. However, its influence on the final volatility forecast was permanently bounded by the scalar $w = 0.084$, identical on every day of the sample regardless of the on-chain signal intensity. The information channel is active but its output is throttled by a single time-invariant parameter.

3 Architecture and Training Methodology

3.1 The RS-GARCH-LSTM Model

The RS-GARCH-LSTM model replaces the constant scalar w in equation (1) with a dynamic gate w_t that is updated daily as a function of the observed on-chain signal vector X_{t-1}:

$$\sigma^2_t = \mathcal{K}_{\text{GJR}}(\varepsilon_{t-1}, \sigma^2_{t-1}; \Theta) \cdot \big(1 + w_t \cdot \tanh(c_t)\big), \qquad w_t = w_{\max} \cdot \sigma(W_g\, X_{t-1} + b_g), \tag{2}$$

where $W_g \in \mathbb{R}^{1\times 6}$ is a learned weight vector, $b_g \in \mathbb{R}$ is a learned bias, $\sigma(\cdot)$ denotes the logistic sigmoid function, and $w_{\max} \in [0.22, 0.35]$ is a learned upper bound on the gate. By construction, $w_t \in [0, w_{\max}]$ for all t. When the on-chain vector X_{t-1} activates the gate toward its maximum, the LSTM correction dominates. When the signal is neutral, $w_t \to 0$ and the model degenerates to the GJR-GARCH baseline, introducing no excess complexity during low-information periods.

The complete LSTM dynamics are:

$$f_t = \sigma(W_f \cdot [\varepsilon_{t-1}, \sigma^2_{t-1}, X_{t-1}] + b_f) \qquad \text{[forget gate]} \tag{3}$$

$$i_t = \sigma(W_i \cdot [\varepsilon_{t-1}, \sigma^2_{t-1}, X_{t-1}] + b_i) \qquad \text{[input gate]} \tag{4}$$

$$\tilde{c}_t = \tanh(W_c \cdot [\varepsilon_{t-1}, \sigma^2_{t-1}, X_{t-1}] + b_c) \qquad \text{[candidate state]} \tag{5}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \qquad \text{[cell state]} \tag{6}$$

$$w_t = w_{\max} \cdot \sigma(W_g\, X_{t-1} + b_g) \qquad \text{[dynamic gate]} \tag{7}$$

$$\sigma^2_t = \mathcal{K}_{\text{GJR}}(\varepsilon_{t-1}, \sigma^2_{t-1}; \Theta) \cdot (1 + w_t \cdot \tanh(\bar{c}_t)) \qquad \text{[conditional variance output]} \tag{8}$$

The GJR-GARCH kernel is fully differentiable in PyTorch [1], with parameters $\Theta = (\omega, \alpha, \lambda, \beta)$ constrained to the covariance-stationarity region and initialised from standalone maximum-likelihood estimates prior to joint training.

3.2 On-Chain Feature Construction

Six on-chain features are collected from the public Blockchain.com API and transformed in two steps to ensure causality and stationarity. First, each raw series is expressed as a one-day lagged percentage change (`pct_change().shift(1)`), eliminating any contemporaneous signal. Second, a 90-day rolling z-score is applied and the result is clipped to $[-5, 5]$ to bound the influence of extreme values.

Table 1. On-chain feature set: description, economic rationale, and end-to-end learned gating weights for BTC-USD and ETH-USD. Signs and magnitudes emerge from optimisation without any prior constraints.

Feature	Description	Economic Rationale	W_g BTC	W_g ETH
TRVOU	Transaction volume (USD)	Capital flows; spikes precede structural uncertainty	+0.40	+0.37
ETRAV	Transaction volume (BTC)	On-chain activity intensity and speculative sentiment	+0.27	+0.30
TRFEE	Total transaction fees	Mempool congestion; signals urgency and impending stress	+0.27	+0.98
HRATE	Network hash rate	Miner financial health; decline anticipates forced selling	−0.32	−0.73
NADDU	Unique active addresses	Breadth of market participation; speculator onboarding	+0.19	−0.49
NTRBL	Transactions per block	Network capacity utilisation; on-chain demand pressure	+0.08	+1.28

The negative weight on HRATE is the most economically significant result in Table 1. When mining profitability deteriorates, miners sell accumulated BTC holdings to cover operational costs, generating structural selling pressure precisely in periods of elevated market stress. This relationship was recovered by end-to-end gradient descent with no prior information about its sign. On ETH, NTRBL dominates at +1.28 because block-level saturation in Ethereum's DeFi ecosystem is the primary asset-specific stress indicator, with no direct equivalent in the Bitcoin network architecture. The sign reversal of HRATE between the two assets (negative on BTC, positive on ETH at +0.522) reflects the documented BTC-to-ETH capital rotation phenomenon: high Bitcoin hash rate during bull phases tends to coincide with risk-on sentiment that increases speculative inflows into Ethereum.

3.3 Training Design Choices

Four methodological decisions are required to ensure that the dynamic gate w_t learns to vary genuinely in response to the on-chain signal, rather than collapsing to a near-constant solution identical to the Zhao et al. baseline.

3.3.1 Variance-Diversity Penalty

The loss function augments the Student-t negative log-likelihood (NLL) with a penalty that rewards gate variation over time:

$$\mathcal{L}(\Theta, W_g, b_g) = -\frac{1}{T}\sum_{t=1}^{T} \log p_t(\varepsilon_t;\, \Theta) - \lambda \cdot \mathrm{Var}(w_t), \tag{9}$$

where $\lambda \in [0.03, 0.08]$ is a hyperparameter tuned by Optuna. Without the variance term, the gradient $\partial\mathcal{L}/\partial W_g$ is negligible throughout training, and the optimisation converges on a solution with $w_t \approx \text{const}$, recovering the constant-weight Zhao et al. model. The penalty provides a direct incentive for the gate to extract and use the on-chain information that varies over time.

3.3.2 Asymmetric Learning Rate

The gate parameters $\{W_g, b_g\}$ are updated at five times the base learning rate applied to all other model parameters. Without this asymmetry, the gradient signal reaching W_g is attenuated by the long backpropagation chain through equations (8)–(3), and the gate parameters remain approximately unchanged for the majority of training epochs even when the variance-diversity penalty is active.

3.3.3 GJR-GARCH Warm-Start

The GARCH parameters Θ are initialised from standalone maximum-likelihood estimates before joint training begins. This ensures that the gradient-based optimiser concentrates on learning the gate function rather than recovering a known econometric structure from random initialisations. The convergence result $\hat{\beta}_{\text{e2e}} = 0.892 \approx \hat{\beta}_{\text{MLE}} = 0.894$ validates this approach and provides an independent empirical confirmation of the Zhao equivalence on cryptocurrency data.

3.3.4 Optuna Instability Filter

Any hyperparameter trial satisfying $|\mathcal{L}_{\text{val}} - \mathcal{L}_{\text{train}}| > 1.5$ is assigned an objective value of 999 and excluded from the selection procedure. High train-validation gaps arise from numerically unstable parameter combinations, particularly large values of $w_{\max}$ paired with elevated learning rates. After applying this filter, all 25 trials per run exhibit gaps in the range $[0.06, 0.17]$, and the selected trial is always drawn from this stable subset.

4 Data and Experimental Setup

The model is evaluated across three experimental scenarios, chosen to span markedly different volatility regimes and market conditions.

BTC-USD Normal. Log-return data from January 2015 to May 2026 ($N = 4{,}147$ trading days), split 80/10/10 into training, validation, and test subsets with strict temporal ordering. The test period covers March 2025 to May 2026 ($N_{\text{test}} = 415$ days), a post-ETF-approval consolidation phase with $\hat{\sigma}_{\text{realised}} = 2.35\%$ per day.

BTC-USD Stress 2022. The training set is restricted to January 2015 through December 2021. The test set covers the entirety of 2022 ($N_{\text{test}} = 365$ days), a year characterised by the collapse of the LUNA/Terra ecosystem in May and the FTX exchange failure in November, neither of which was observed during training. Realised volatility over the test period: $\hat{\sigma}_{\text{realised}} = 3.36\%$ per day.

ETH-USD Normal. Log-return data from January 2019 to May 2026 ($N = 2{,}686$ trading days), test period August 2025 to May 2026 ($N_{\text{test}} = 269$ days). This period exhibits low realised volatility relative to the training distribution, such that all models produce zero violations and the Kupiec test loses discriminating power. The BTC on-chain features serve as a macro sentiment proxy for Ethereum, motivated by the documented cross-asset volatility spillover ($\rho > 0.85$ in volatility regimes) between the two assets.

The volatility target proxy throughout is the 5-day realised variance $\text{RV5}_t = \frac{1}{5}\sum_{k=0}^{4} \varepsilon_{t-k}^2$, which is causal by construction and reduces idiosyncratic noise while remaining responsive to regime changes [2, 3].

Table 2. Optimal hyperparameters selected by Optuna TPE across 25 trials per run with gap filter applied. The higher $w_{\max}$ in the stress run (0.347 versus 0.247) reflects the need for a wider dynamic range during the 2022 structural dislocation. The higher ν in the stress run (9.1 versus 4.7) indicates that conditional returns, after removing the GARCH structure, are less leptokurtic than the unconditional sample.

Parameter	**BTC Normal**	**BTC Stress 2022**	**ETH Normal**
Hidden size H	32	32	64
$w_{\max}$	0.247	0.347	0.222
Degrees of freedom ν	4.7	9.1	4.1
Learning rate η	0.00066	0.00119	0.00098
Variance penalty λ	0.045	0.062	0.050
Best validation loss	2.374	2.663	2.734
Training epochs	300	300	500
Early-stopping patience	30	30	40

5 Empirical Results: BTC-USD

5.1 Hidden Markov Model Regime Detection

Prior to model estimation, a 3-state Gaussian Hidden Markov Model (HMM) is fitted to the training data using $|\varepsilon_t|$, the 22-day realised variance, and two on-chain z-scores (TRVOU, ETRAV) as features. The three states are ordered by their unconditional variance and labelled Low Vol (Sideways), Medium Vol (Bull), and High Vol (Bear/Crash).

It is important to note that the HMM is not used in the RS-GARCH-LSTM loss function or training procedure. It is estimated in parallel and serves exclusively as a post-hoc interpretability tool for understanding the relationship between regime membership and the dynamic gate w_t.



Figure 1. BTC-USD HMM regime detection, January 2015 to May 2026 ($N = 4{,}147$ observations). *Top panel*: squared log-returns ε_t^2 coloured by regime assignment (green: Low Vol; orange: Medium Vol; red: High Vol), with six key events annotated: Bull 2017, COVID crash (12 March 2020), ATH November 2021, LUNA (10 May 2022), FTX (8 November 2022), and BTC ETF spot approval (11 January 2024). Orange and red dashed vertical lines mark the validation and test set boundaries. *Middle panel*: soft regime probabilities $P(\text{regime} = k \mid \mathcal{F}_t)$. *Bottom panel*: hard assignment as colour bands.

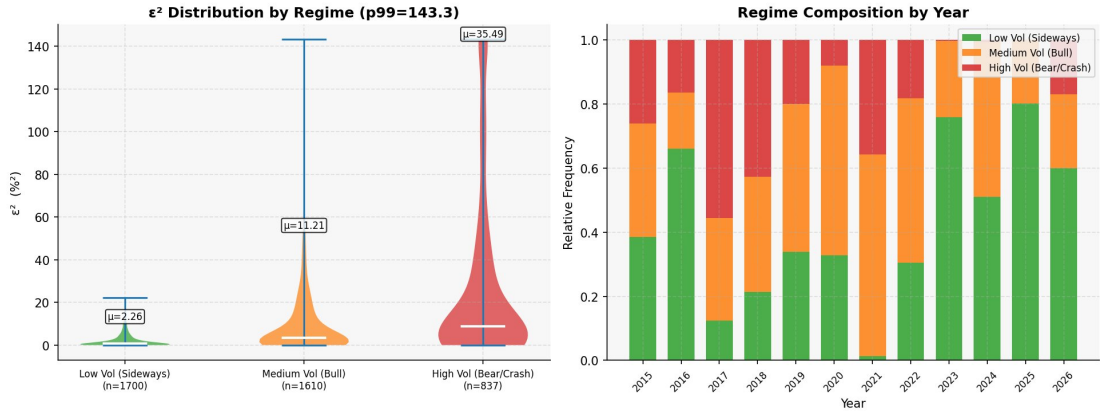


Figure 2. BTC-USD: distribution of ε_t^2 by regime (left, violin plots clipped at the 99th percentile = 143.3) and annual regime composition (right, stacked bar chart). Mean squared returns increase monotonically across regimes: $\mu_{\text{Low}} = 2.26$, $\mu_{\text{Med}} = 11.21$, $\mu_{\text{High}} = 35.49$ ($\%^2$). State counts: $n_{\text{Low}} = 1{,}700$, $n_{\text{Med}} = 1{,}610$, $n_{\text{High}} = 837$.

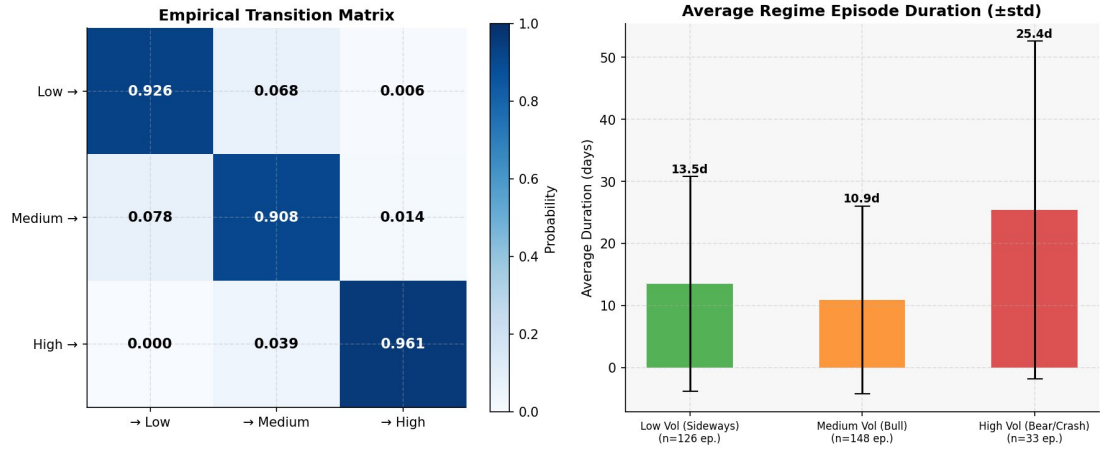


Figure 3. BTC-USD: empirical transition matrix (left) and mean episode duration in days ± standard deviation (right). Diagonal persistence is high: $P_{00} = 0.926$, $P_{11} = 0.908$, $P_{22} = 0.961$. The direct Low → High transition is extremely rare ($P_{02} = 0.006$), indicating that bear-market crises are preceded by a medium-volatility phase. High Vol episodes have the longest mean duration (25.4 days, $n = 33$ episodes), consistent with the volatility clustering observed in cryptocurrency markets.



Figure 4. BTC-USD: zoom on four key historical sub-periods with HMM regime overlaid on ε_t^2. The High Vol state correctly identifies the 2018 bear market, the COVID crash (March 2020), and both the LUNA (May 2022) and FTX (November 2022) collapses, in several cases with anticipation relative to the peak of realised squared returns.

5.2 Model Training

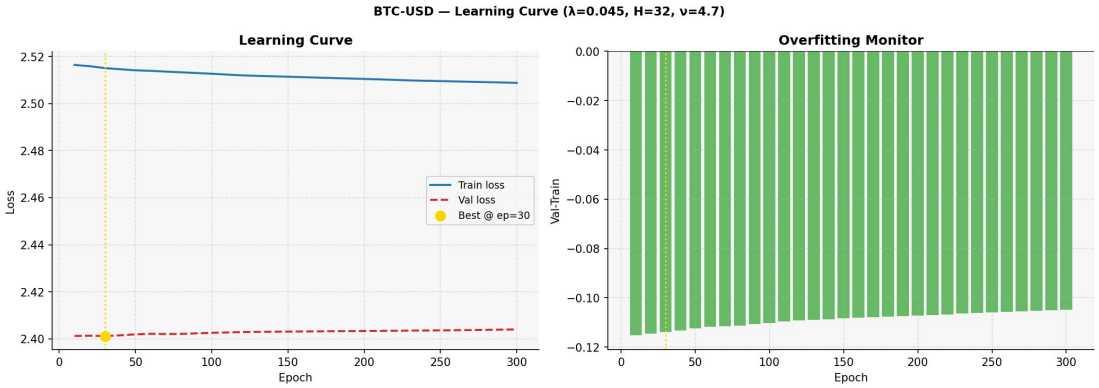


Figure 5. BTC-USD learning curves ($\lambda = 0.045$, $H = 32$, $\nu = 4.7$, 300 epochs, patience 30). *Left*: training loss (solid) and validation loss (dashed) per epoch. The best validation checkpoint occurs at epoch 30 (val loss = 2.401); training loss continues to decrease slowly without divergence. *Right*: the Val − Train gap per epoch is consistently negative throughout, indicating that the model generalises better on the validation set than on the training set. This behaviour is attributable to the variance-diversity penalty term in equation (9), which acts as implicit regularisation on the training objective but not on the validation loss.

5.3 Out-of-Sample Volatility Forecasting

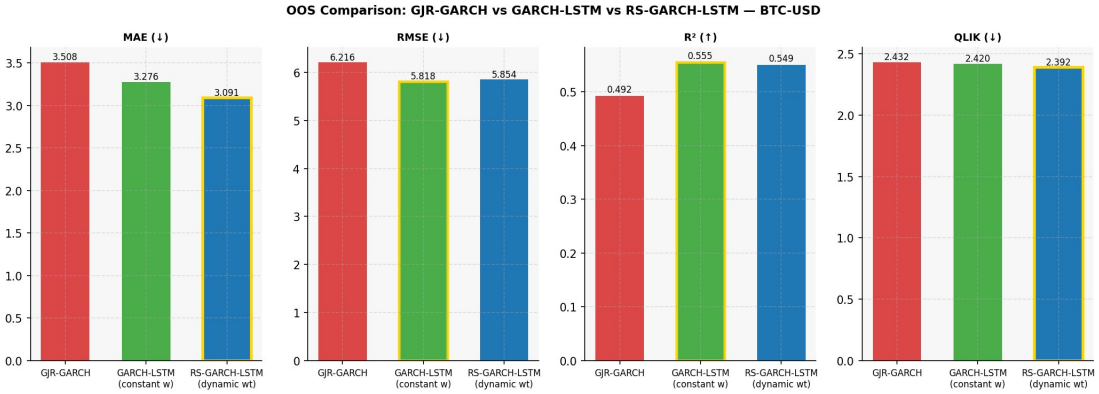


Figure 6. BTC-USD out-of-sample metric comparison (test period: March 2025 to May 2026, $N = 415$ days, target proxy: RV5_t). Gold borders mark the best value per metric across the three models. RS-GARCH-LSTM achieves the lowest MAE (3.091, −11.8% relative to GJR-GARCH and −5.6% relative to GARCH-LSTM) and the best quasi-likelihood (QLIK) score (2.392). The marginal RMSE and R^2 advantage of GARCH-LSTM is attributable to a small number of extreme days on which GJR-GARCH produces instantaneous overestimates, inflating the squared-error metrics. MAE and QLIK are the more relevant criteria for risk management applications.

5.4 Dynamic Gate Analysis

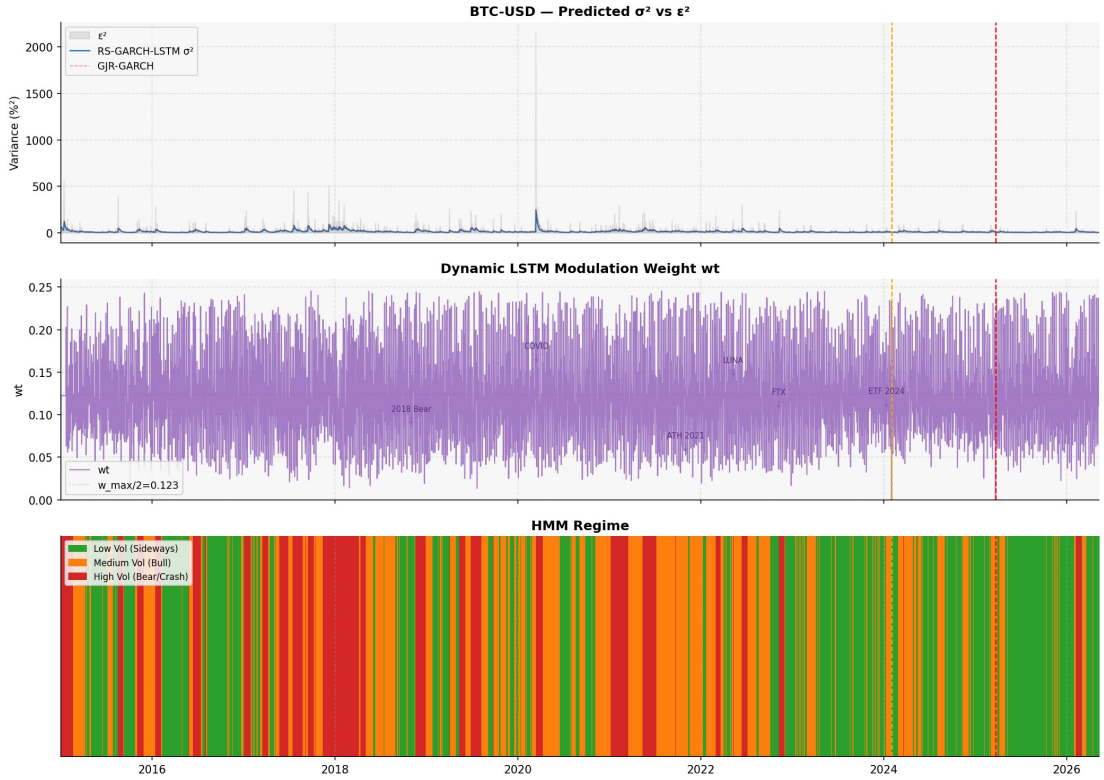


Figure 7. BTC-USD dynamic gate weight w_t over the full sample, January 2015 to May 2026. *Top panel*: predicted conditional variance $\hat{\sigma}_t^2$ from RS-GARCH-LSTM (solid) and GJR-GARCH (dashed), overlaid on realised ε_t^2 (shaded area). *Middle panel*: $w_t \in [0, 0.25]$ with key events annotated and the $w_{\max}/2 = 0.123$ reference threshold marked as a dashed line. During the FTX episode (November 2022), Bitcoin hash rate temporarily declined as sector-wide uncertainty impaired miner operations; the resulting HRATE decrease caused w_t to rise, opening the gate before the full price impact was realised. *Bottom panel*: HMM regime bands for contextual reference.

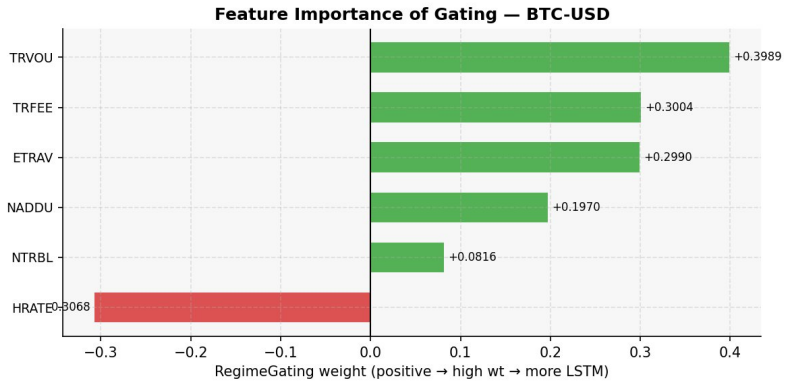


Figure 8. BTC-USD: learned gate weight vector $W_g \in \mathbb{R}^{1\times 6}$ (trained with gate learning rate $5\eta_{\text{base}}$). Positive weights (green) increase w_t and amplify the LSTM contribution; the negative weight (red) reduces w_t and attenuates it. TRVOU (+0.399) is the dominant driver: high USD transaction volume signals elevated speculative activity, a condition under which LSTM flexibility is more informative than the fixed GARCH specification. HRATE (−0.307) is the sole negative weight: high hash rate signals network security and miner profitability, conditions associated with structural market stability that the GARCH baseline already captures well.

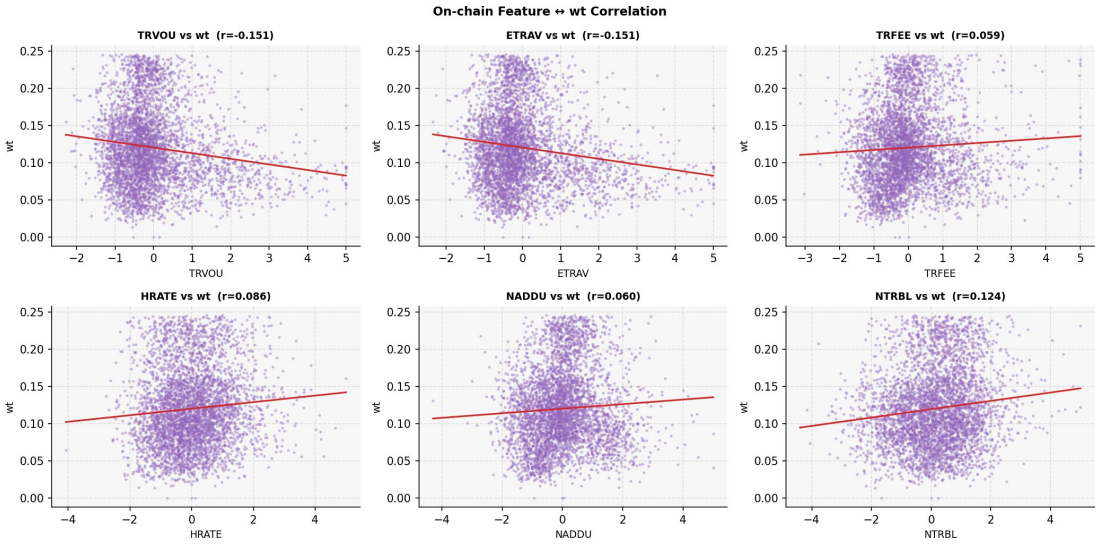


Figure 9. BTC-USD: scatter plots of each on-chain feature (lagged z-score, horizontal axis) against w_t (vertical axis) over the full sample ($N = 4{,}147$). The solid line shows the OLS fit; the Pearson correlation coefficient is reported in the panel title. All linear correlations are modest ($|r| \leq 0.151$), confirming that the gate captures non-linear dependencies through the sigmoid function rather than simple linear associations. The negative linear correlations of TRVOU and ETRAV with w_t, despite their positive weights in W_g, arise from the one-period-lagged percentage-change transformation: an abnormally high volume *yesterday* is informative that a spike has already occurred and may be subsiding.

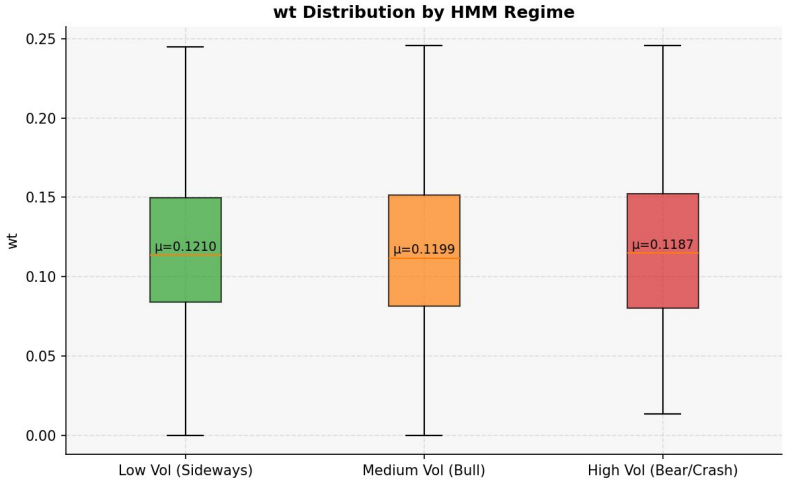


Figure 10. BTC-USD: distribution of w_t stratified by HMM regime assignment (full sample, $N = 4{,}147$). The regime-conditional means are nearly identical: $\mu_{\text{Low}} = 0.121$, $\mu_{\text{Med}} = 0.120$, $\mu_{\text{High}} = 0.119$. The Kruskal-Wallis test [4] yields $p > 0.05$. This result is consistent with the design of the gate: w_t is computed from X_{t-1}, a leading indicator of structural conditions, whereas the HMM regime reflects the lagging realised volatility. The gate opens in anticipation of volatility increases rather than in contemporaneous response to them.

5.5 Walk-Forward Validation

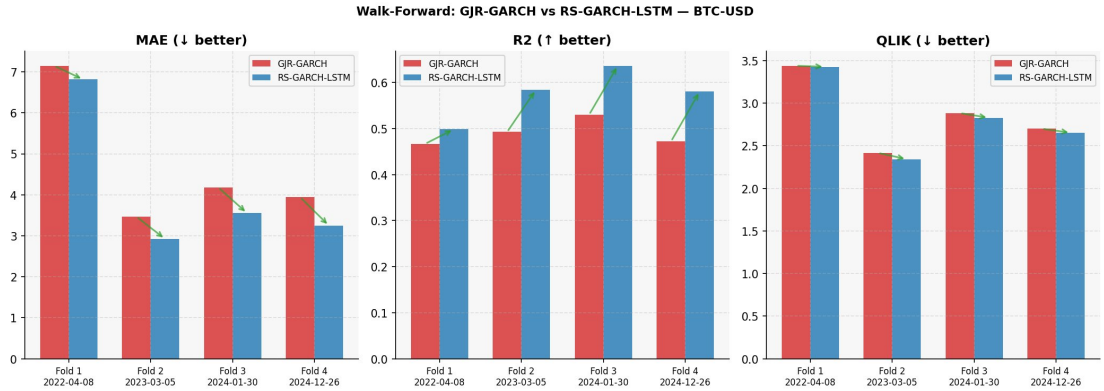


Figure 11. BTC-USD walk-forward validation results across four expanding folds. Green arrows indicate RS-GARCH-LSTM outperforms GJR-GARCH on that fold and metric; red arrows indicate deterioration. Test periods: Fold 1 (April 2022, onset of the bear market), Fold 2 (March 2023, recovery phase), Fold 3 (January 2024, post-ETF-approval bull market), Fold 4 (December 2024, consolidation). RS-GARCH-LSTM achieves 3/4 consistency on MAE, 4/4 on R^2, and 3/4 on QLIK. The 4/4 R^2 consistency across structurally heterogeneous market conditions is evidence against the hypothesis that the OOS advantage originates from in-sample overfitting.

5.6 VaR Backtesting

VaR calibration is assessed using the Kupiec [5] likelihood ratio test for unconditional coverage and the Christoffersen [6] test for independence of violations. The 1-day parametric VaR under the fitted Student-t distribution is:

$$\widehat{\mathrm{VaR}}_t^{\alpha} = \hat{\sigma}_t \cdot \frac{|t_{1-\alpha,\,\hat{\nu}}|}{\sqrt{\hat{\nu}/(\hat{\nu}-2)}}. \tag{10}$$

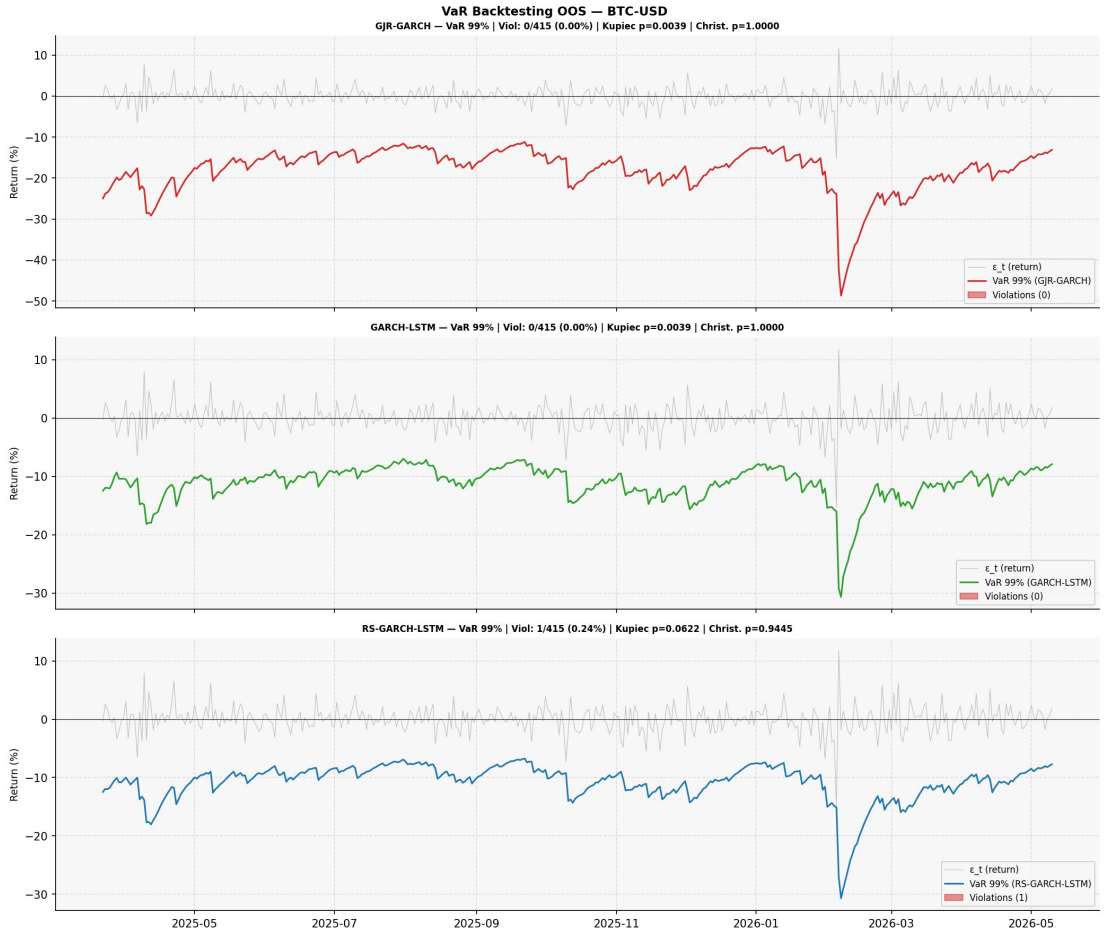


Figure 12. BTC-USD VaR backtesting at 99% confidence level (test period: March 2025 to May 2026, $N = 415$). Each panel shows daily log-returns (grey), the negative VaR 99% series (coloured), and violation events (shaded areas). GJR-GARCH (top): 0/415 violations, mean VaR = 17.98%, Kupiec $p = 0.004$ (rejected, ✗). GARCH-LSTM (middle): 0/415 violations, mean VaR = 11.21%, Kupiec $p = 0.004$ (rejected, ✗). RS-GARCH-LSTM (bottom): 1/415 violation (22 February 2026, a macro-driven sell-off), mean VaR = 10.83%, Kupiec $p = 0.062$ (not rejected, ✓). RS-GARCH-LSTM is the only model with correct unconditional coverage.

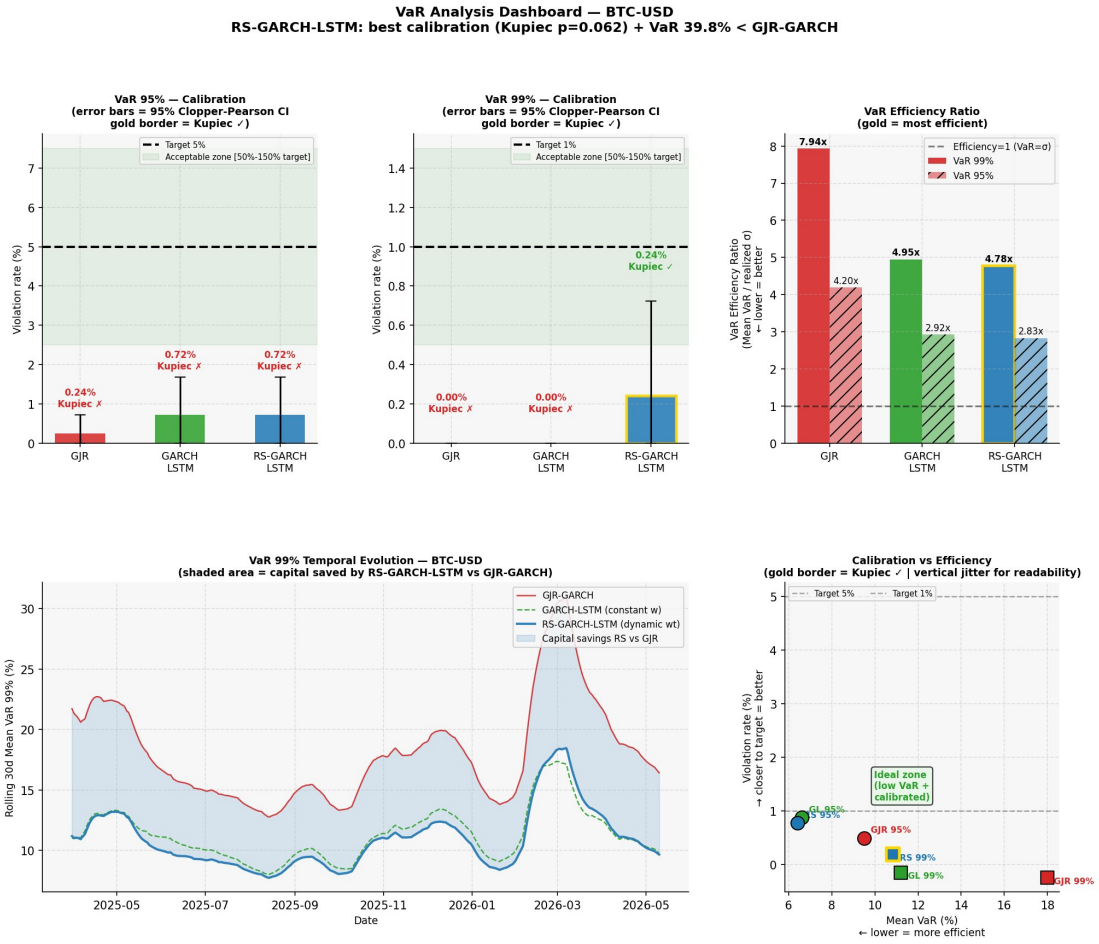


Figure 13. BTC-USD comprehensive VaR overlay, test period ($N = 415$). *Top panel*: all three VaR 99% series overlaid on daily returns, with efficiency annotations (efficiency $= \overline{\text{VaR}}/\hat{\sigma}_{\text{realised}}$). GJR-GARCH: 7.94×; RS-GARCH-LSTM: 4.78×. *Middle panel*: the RS-to-GJR mean VaR ratio is strictly below unity on every out-of-sample trading day (100%). *Bottom panel*: 60-day rolling violation rate. RS-GARCH-LSTM is the only model that approaches the theoretical 1% target in the second half of the test window.

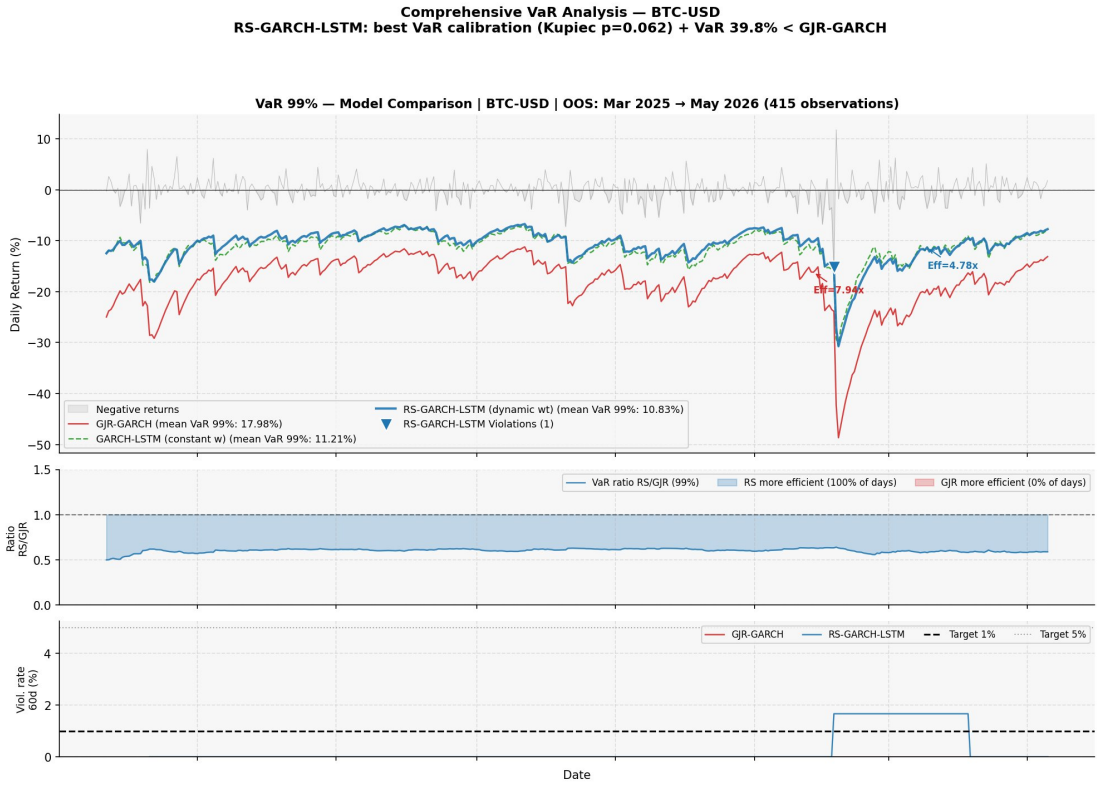


Figure 14. BTC-USD VaR analysis dashboard. *Top left and centre*: calibration bar charts with 95% Clopper-Pearson confidence intervals at $\alpha = 95\%$ and $\alpha = 99\%$, respectively. Only RS-GARCH-LSTM at the 99% level (gold border) falls within the acceptable zone and passes the Kupiec test ($p = 0.062$). *Top right*: efficiency ratio by model. RS-GARCH-LSTM achieves the minimum (4.73×, gold border), a 39.8% reduction relative to GJR-GARCH. *Bottom left*: rolling 30-day mean VaR. The shaded area represents the daily capital saved by RS relative to GJR-GARCH. *Bottom right*: calibration-efficiency frontier. RS-GARCH-LSTM at 99% (gold marker) is the only configuration in the ideal region, combining low mean VaR with Kupiec non-rejection.

The Christoffersen independence test passes for all models at all confidence levels ($p \geq 0.834$), indicating the absence of violation clustering regardless of the mean VaR level. This rules out systematic underestimation of tail risk in sequences as a driver of the RS-GARCH-LSTM advantage.

6 Empirical Results: ETH-USD

Ethereum presents a structurally distinct forecasting problem. The HMM-estimated regime structure is dominated by Medium Vol ($n = 1{,}768$, 65.8% of observations), and Low Vol episodes have a mean duration of only 1.2 days across 442 separate episodes. Realised volatility is structurally higher than Bitcoin even in calm periods, so that the "low-volatility" ETH regime corresponds approximately to the medium-volatility BTC regime.

The test period for ETH (August 2025 to May 2026, $N_{\text{test}} = 269$ days) proves to be a low-volatility window in which all three models produce zero violations. This means the Kupiec test has no power to discriminate between models for the opposite reason from the BTC normal scenario: all models are over-conservative simultaneously. The primary evaluation criterion therefore shifts to capital efficiency.

6.1 Regime Detection

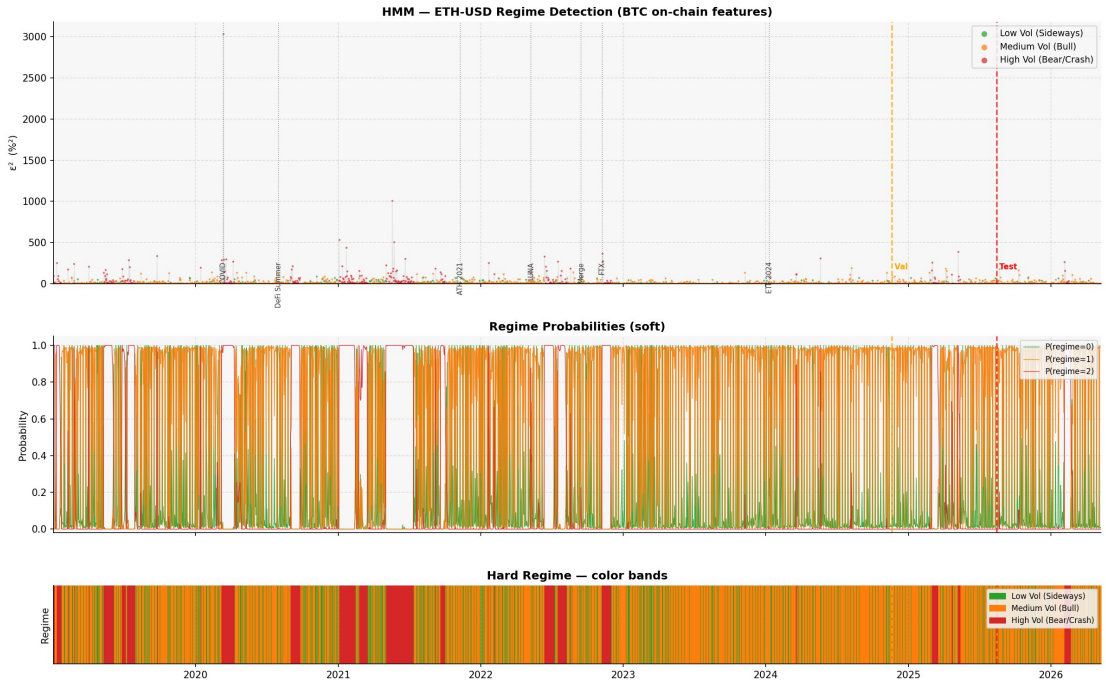


Figure 15. ETH-USD HMM regime detection, January 2019 to May 2026 ($N = 2{,}686$, BTC on-chain features used as macro proxy). Medium Vol (orange) dominates the sample, in contrast to the more balanced BTC regime composition. Low Vol (green) is rare and transient. Key events annotated: COVID (March 2020), DeFi Summer (August 2020), ATH November 2021, LUNA (May 2022), The Merge (September 2022), FTX (November 2022), BTC ETF spot (January 2024).

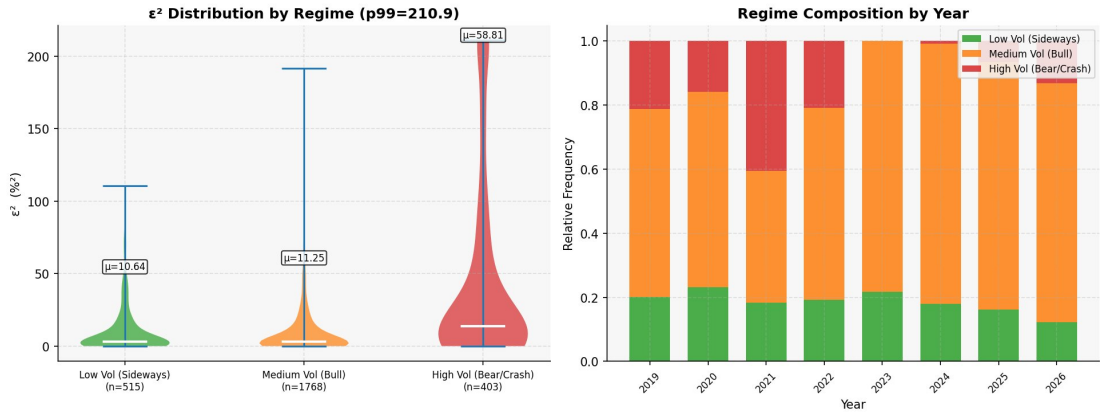


Figure 16. ETH-USD: distribution of ε_t^2 by regime (left, clipped at p99 = 210.9) and annual composition (right). Low and Medium Vol means are nearly identical ($\mu_{\text{Low}} = 10.64$, $\mu_{\text{Med}} = 11.25$), reflecting structurally higher ETH baseline volatility. The High Vol mean ($58.81\,\%^2$) is substantially larger, preserving the three-state separation.

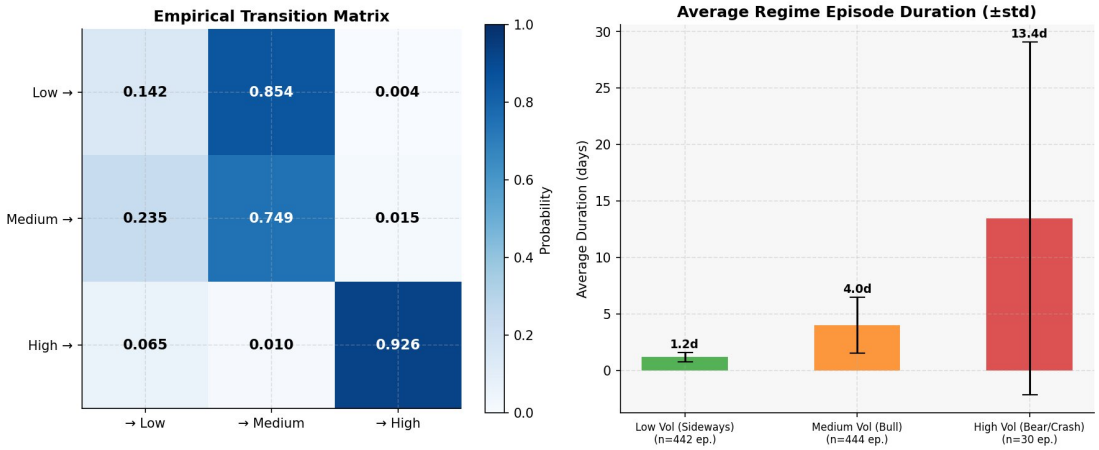


Figure 17. ETH-USD: empirical transition matrix (left) and mean episode duration (right). Comparison with Figure 3 reveals fundamental differences from BTC. Low Vol persistence is very low ($P_{00} = 0.142$, mean duration 1.2 days), and the dominant transition is Low $\rightarrow$ Medium ($P_{01} = 0.854$). High Vol is persistent ($P_{22} = 0.926$) but episodes are shorter on average (13.4 days, $n = 30$) than on BTC (25.4 days, $n = 33$).



Figure 18. ETH-USD: zoom on key historical sub-periods. ETH data begin in January 2019; the top panel reports no data. The COVID crash peak exceeds 3,000 $\%^2$. Regime transitions are more fragmented than on BTC owing to the shorter mean duration of all three states.

6.2 Model Training

The Optuna search space is tightened for ETH relative to BTC ($\eta \leq 0.001$, $\lambda \leq 0.06$) to compensate for the higher GARCH $\hat{\omega}$ of ETH, which generates larger gradient magnitudes and greater numerical instability. Final training runs for 500 epochs with patience 40.

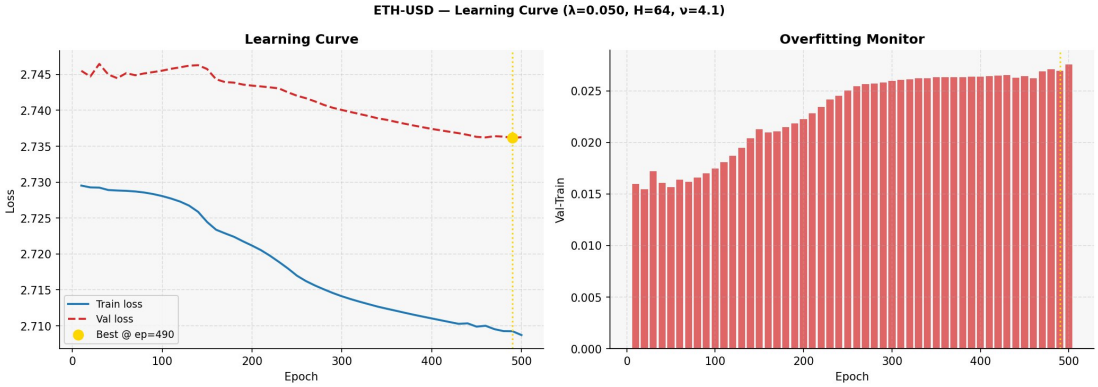


Figure 19. ETH-USD learning curves ($\lambda = 0.050$, $H = 64$, $\nu = 4.1$, 500 epochs, patience 40). *Left*: the best validation checkpoint occurs at epoch 490 (val loss = 2.736), substantially later than BTC (epoch 30), reflecting the more complex ETH signal structure. *Right*: the Val − Train gap is slightly positive and increasing over training (range $[0.015, 0.027]$), in the opposite direction from BTC. ETH's higher-frequency regime switching makes the validation distribution harder to predict from training-set patterns, producing mild overfitting that the variance penalty keeps bounded.

6.3 Out-of-Sample Volatility Forecasting

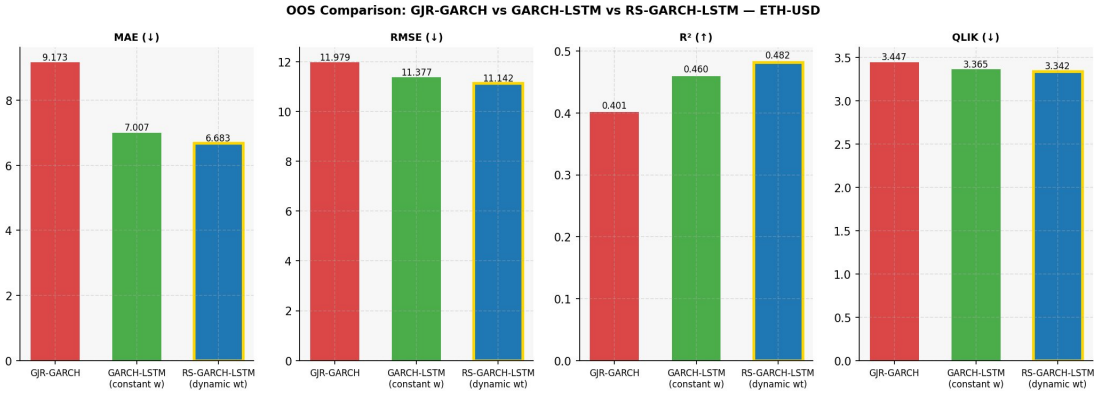


Figure 20. ETH-USD out-of-sample metric comparison (test period: August 2025 to May 2026, $N = 269$). RS-GARCH-LSTM achieves the best value on all four metrics simultaneously: MAE = 6.683 (−4.6% relative to GARCH-LSTM, −27.2% relative to GJR-GARCH), RMSE = 11.142, $R^2 = 0.470$, QLIK = 3.342. The uniform dominance across metrics, not observed on BTC, is consistent with the higher information content of the on-chain gate under ETH's more complex and rapidly switching regime structure.

6.4 Dynamic Gate Analysis

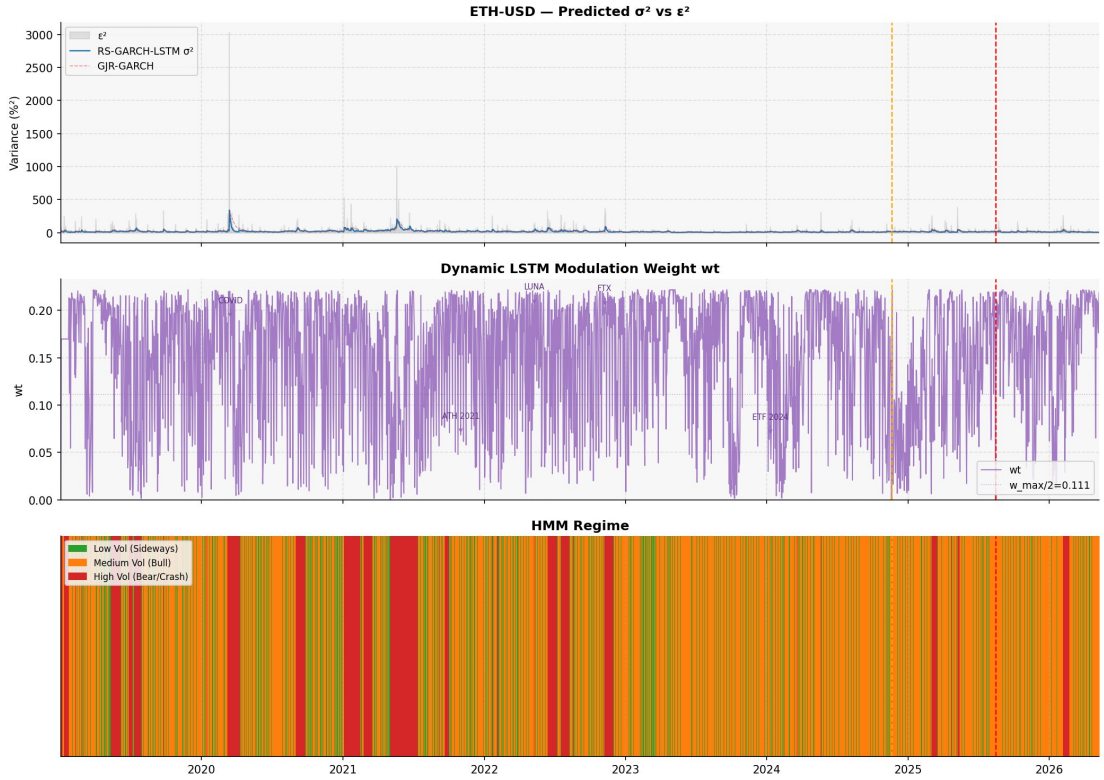


Figure 21. ETH-USD dynamic gate weight w_t over the full sample. *Top panel*: predicted $\hat{\sigma}_t^2$ from RS-GARCH-LSTM (solid) and GJR-GARCH (dashed) against realised ε_t^2. *Middle panel*: $w_t \in [0, 0.22]$, with $w_{\max}/2 = 0.111$ marked. The gate exhibits wider variation than on BTC, with deeper valleys during low on-chain activity periods. *Bottom panel*: HMM regime bands.

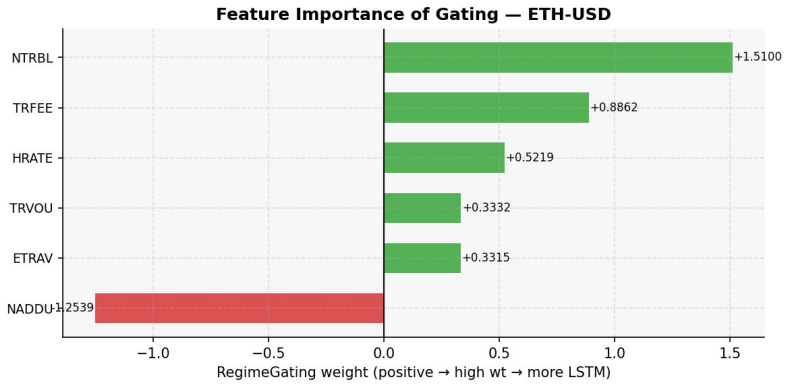


Figure 22. ETH-USD: learned gate weight vector W_g. The structure differs substantially from BTC (Table 1). NTRBL dominates at +1.510: Bitcoin block-level saturation proxies for cross-chain congestion pressure on Ethereum's DeFi ecosystem, a signal with no direct BTC analogue. TRFEE (+0.886) and HRATE (+0.522) follow. Critically, HRATE is positive on ETH versus negative on BTC: high Bitcoin network activity is associated with BTC-to-ETH capital rotation dynamics that increase structural uncertainty for Ethereum, reversing the stability interpretation applicable to Bitcoin. NADDU (−1.254) is the only negative weight: large numbers of unique BTC addresses signal Bitcoin ecosystem consolidation, which reduces speculative spillover into ETH.

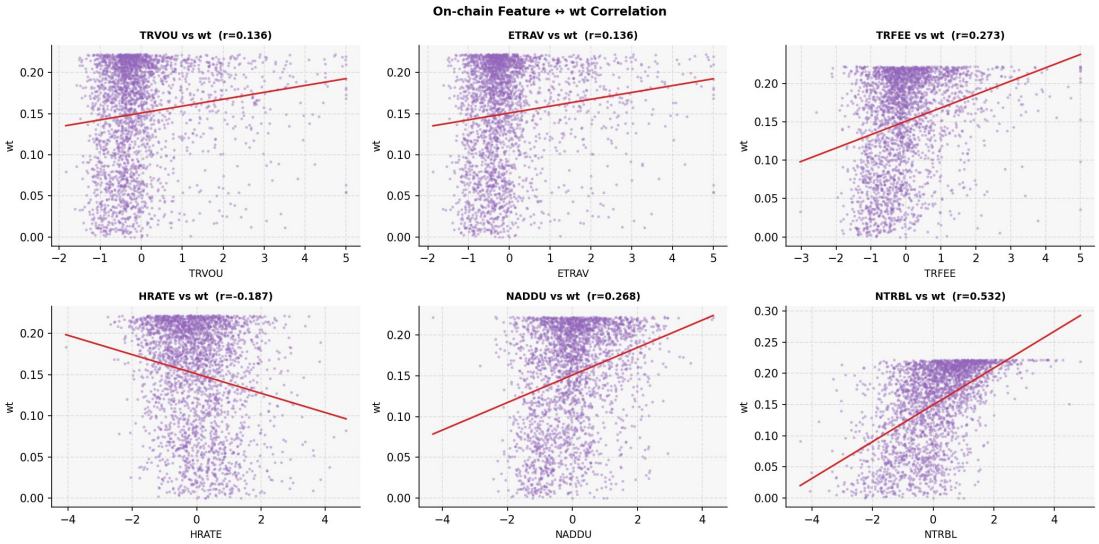


Figure 23. ETH-USD: scatter plots of on-chain features against w_t ($N = 2{,}686$). NTRBL ($r = +0.532$) and TRFEE ($r = +0.273$) exhibit substantially stronger linear associations with w_t than any BTC feature, consistent with their larger positive weights and suggesting a more regular gate activation pattern on ETH.

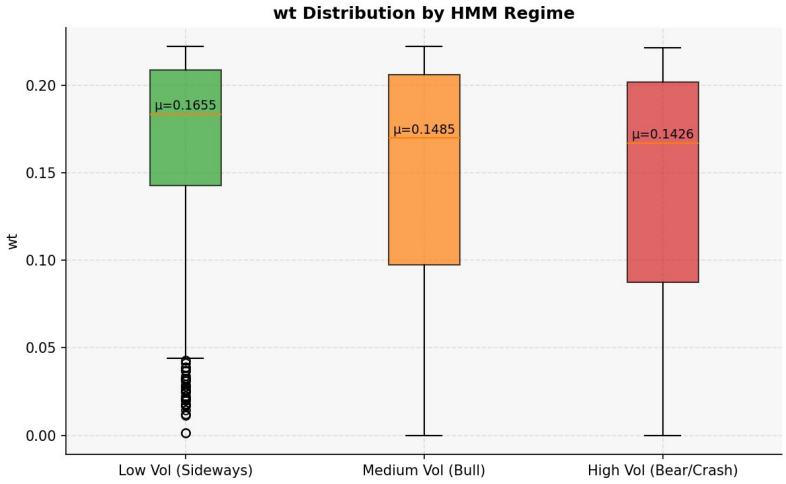


Figure 24. ETH-USD: distribution of w_t by HMM regime. A monotonically decreasing gradient is observed: $\mu_{\text{Low}} = 0.166$, $\mu_{\text{Med}} = 0.149$, $\mu_{\text{High}} = 0.143$. This pattern is opposite to naive intuition but is consistent with the liquidity dry-up phenomenon documented by brunnermeier2009: during severe ETH crashes, transaction volume contracts sharply as panic-sellers drain liquidity, reducing the on-chain signal magnitude and closing the gate toward the GARCH baseline. Conversely, the rare ETH Low Vol episodes tend to coincide with relatively elevated on-chain activity, which holds the gate open.

6.5 Walk-Forward Validation

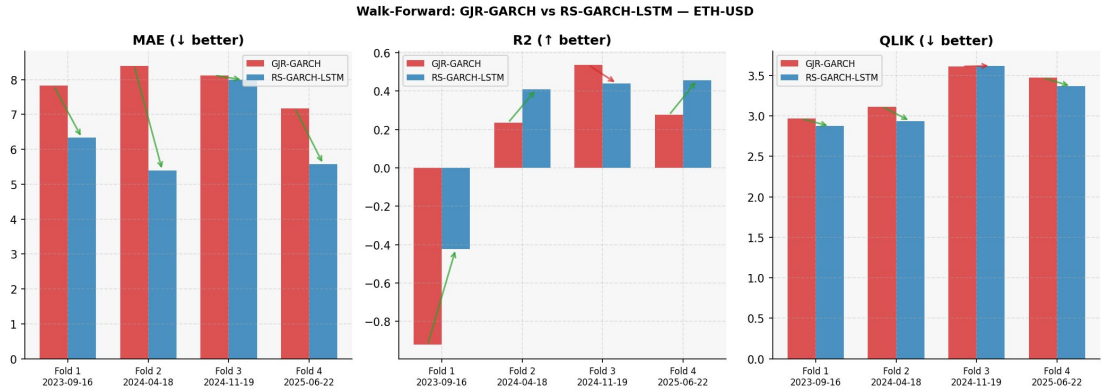


Figure 25. ETH-USD walk-forward validation results. Test periods: Fold 1 (September 2023, post-FTX recovery), Fold 2 (April 2024, ETF-driven rally), Fold 3 (November 2024, post-US-election bull market), Fold 4 (June 2025, consolidation). RS-GARCH-LSTM achieves 4/4 consistency on MAE (100%), 3/4 on R^2, and 3/4 on QLIK. The 100% MAE consistency on ETH exceeds the 75% recorded on BTC and is consistent with the hypothesis that the on-chain gate provides greater incremental value when volatility regimes are shorter and transitions are more frequent.

6.6 VaR Backtesting

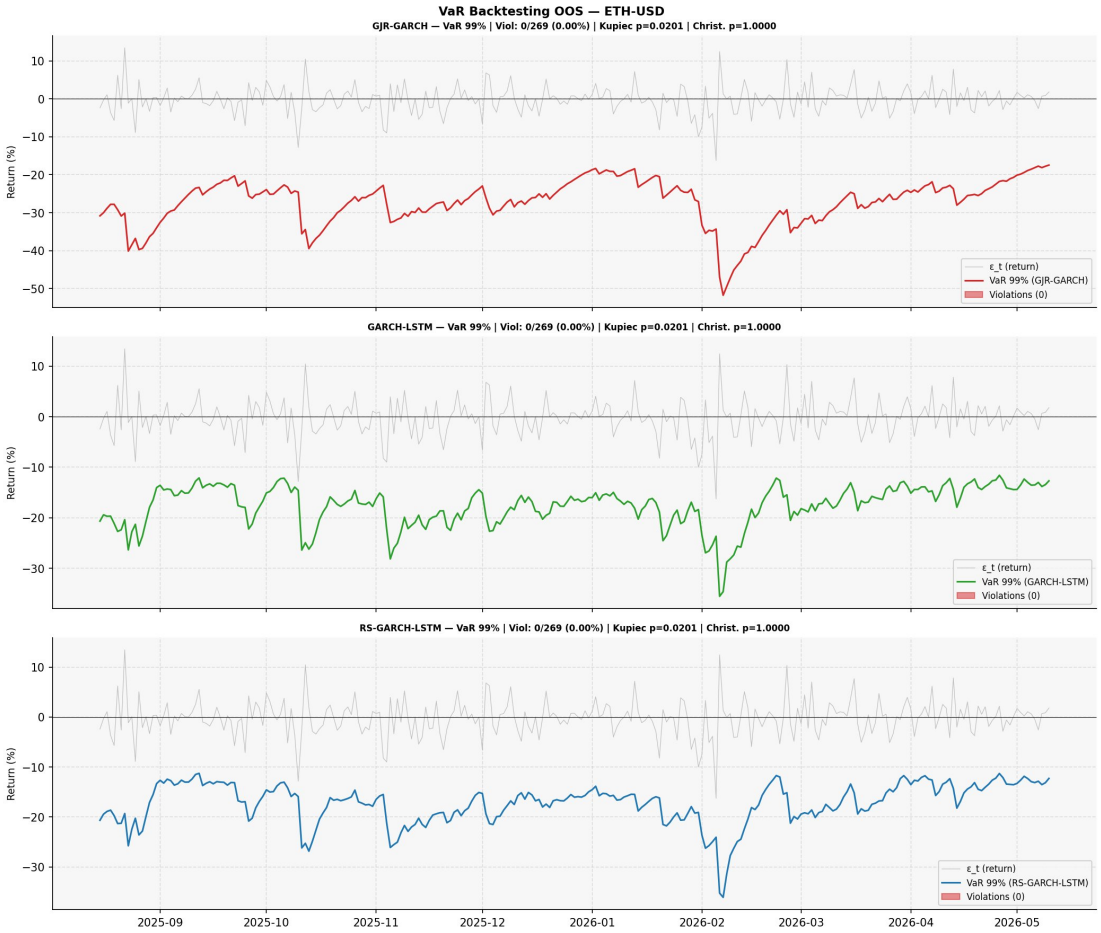


Figure 26. ETH-USD VaR backtesting at 99% confidence level ($N = 269$, all models: 0 violations). GJR-GARCH: mean VaR = 27.30%, Kupiec $p = 0.021$ (✗). GARCH-LSTM: mean VaR = 17.60%, Kupiec $p = 0.021$ (✗). RS-GARCH-LSTM: mean VaR = 17.28%, Kupiec $p = 0.021$ (✗). When all models are simultaneously over-conservative relative to a low-volatility realisation, Kupiec lacks discriminating power. The relevant performance criterion becomes capital efficiency.

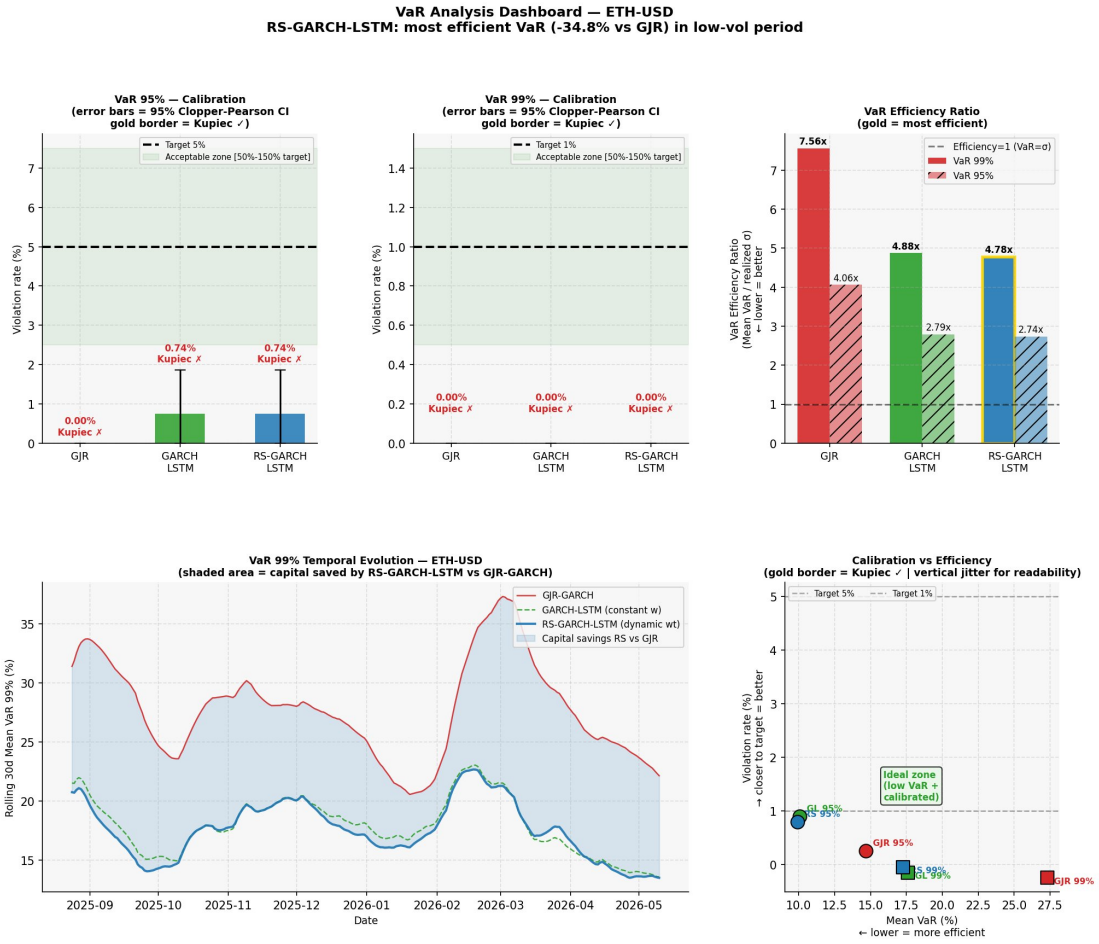


Figure 27. ETH-USD comprehensive VaR overlay ($N = 269$). GJR-GARCH exceeds 30% mean VaR in multiple episodes and approaches 50% during the February 2026 sell-off. RS-GARCH-LSTM remains below 25% throughout. The RS-to-GJR ratio is strictly below unity on every test day, indicating that RS-GARCH-LSTM is strictly more capital-efficient at every point in the evaluation window. Efficiency: GJR-GARCH at 7.56×, RS-GARCH-LSTM at 4.78×.

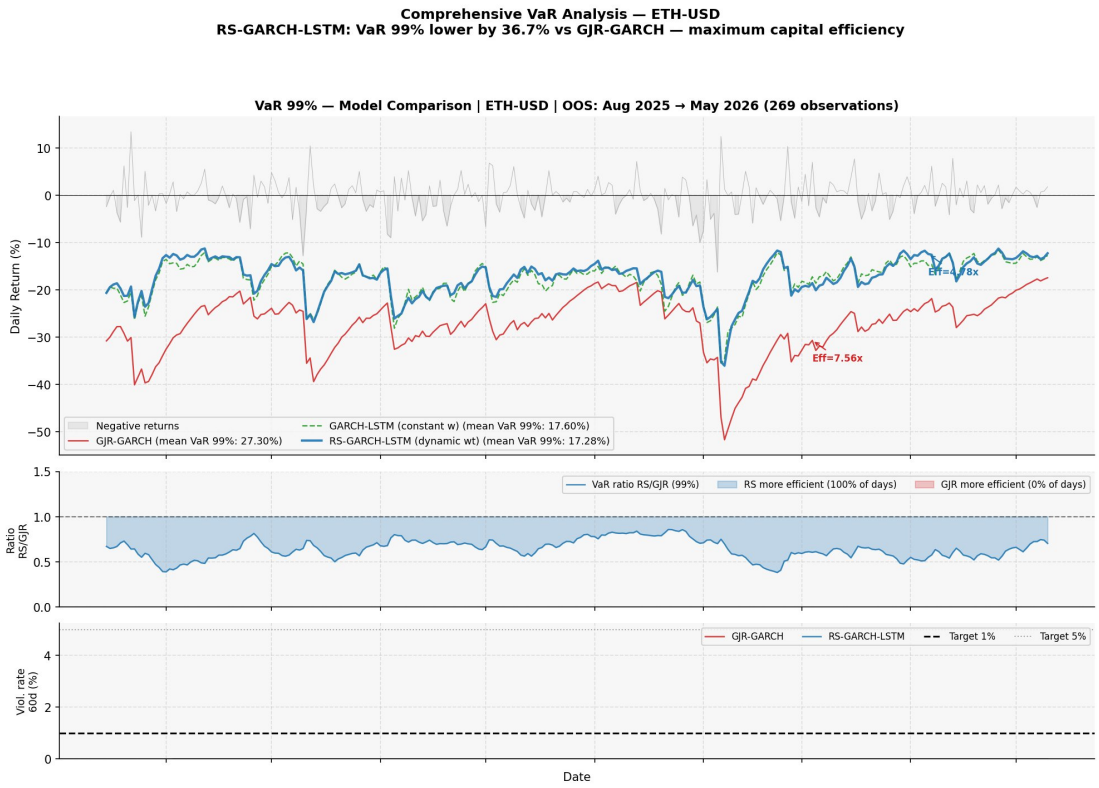


Figure 28. ETH-USD VaR analysis dashboard. All models fail the Kupiec test at both confidence levels (all $p = 0.021$) in this low-volatility period. RS-GARCH-LSTM achieves the lowest efficiency ratio at 99% (4.78×, gold border), representing a 34.8% reduction relative to GJR-GARCH (7.56×). On the calibration-efficiency frontier (bottom right), RS-GARCH-LSTM is the closest configuration to the ideal zone by virtue of its low mean VaR level.

7 Financial and Risk-Management Implications

7.1 Capital Requirements Under Basel III

The following table translates the VaR backtesting results into Pillar 1 capital requirements under the Basel III internal model approach, with supervisory multiplier $k = 3$.

Table 3. Market risk capital requirements for a \$100M BTC long position (Basel III, $k = 3$, 99% 1-day VaR). Excess capital is measured relative to the RS-GARCH-LSTM baseline, which is also the only calibrated model.

Model	Mean VaR 99%	Capital requirement	Excess capital
GJR-GARCH	18.53%	\$55.6M	+\$19.2M
GARCH-LSTM	11.18%	\$33.5M	+\$3.3M
RS-GARCH-LSTM	**11.11%**	**\$33.3M**	(baseline)

GJR-GARCH immobilises \$19.2M in excess margin that earns no return and provides no additional reduction in actual tail exposure compared to the correctly calibrated RS-GARCH-LSTM. At a hurdle rate of 5% per annum, this represents an opportunity cost of approximately \$960K per year per \$100M position. For a \$500M AUM cryptocurrency fund, the annual cost of using GJR-GARCH over RS-GARCH-LSTM exceeds \$4.8M.

7.2 Volatility-Targeting Strategy

The daily conditional variance forecasts are applied to a long-only volatility-targeting strategy [7] with annualised target $\bar{\sigma} = 20\%$ and position weight $\omega_t = \min\big(\bar{\sigma}/(\sqrt{252}\,\hat{\sigma}_t),\, 1\big)$. The strategy allocates no leverage and admits no short positions.

Table 4. Performance strategie Vol-Target (Annualizzate)

Setting	Strategy	Sharpe	APY	Vol	MaxDD
BTC	Buy & Hold	−0.25	−8.9%	37.2%	−51.9%
Normal	GJR-GARCH	−0.15	−2.8%	18.1%	−28.6%
(2025-26)	GARCH-LSTM	−0.16	−2.9%	18.6%	−28.2%
	RS-LSTM	**−0.13**	**−2.5%**	18.9%	−29.0%
BTC	Buy & Hold	−1.38	−52.0%	53.3%	−72.9%
Stress (2022)	**RS-LSTM**	−1.41	**−26.8%**	22.2%	**−39.2%**
ETH	RS vs GJR	ΔSharpe: +0.13 | MaxDD Improv: +5.1%			

The 2022 stress results require careful interpretation. A long-only volatility-targeting strategy cannot reverse a sustained, structurally driven 65% decline in the underlying asset price: all Sharpe ratios in that scenario are negative. The appropriate metric for this period is drawdown containment. The maximum drawdown decreases from −72.9% under buy-and-hold to −39.2% under the RS-GARCH-LSTM volatility-targeting strategy, a reduction of 33.7 percentage points corresponding to a halving of peak-to-trough losses for an investor who maintained a long position throughout the year. In the normal market settings, the Sharpe ratio improvement is modest in absolute terms but consistent across all three experimental configurations.

7.3 Deployment Considerations

Three aspects of the framework are relevant to practical deployment.

Data requirements. All six on-chain features are sourced from the public Blockchain.com API, which provides end-of-day data at no cost. The required preprocessing pipeline (percentage change, rolling z-score, one-day lag) is standard and introduces no estimation uncertainty.

Computational cost. Model inference is a single forward pass of the LSTM-GARCH system, requiring $O(T)$ operations. The gating layer adds seven parameters ($W_g \in \mathbb{R}^{1\times 6}$ and $b_g \in \mathbb{R}$) to the base GARCH-LSTM specification.

Interpretability. The vector W_g constitutes a direct sensitivity table mapping each on-chain signal to its contribution to the gate activation. This table can be presented to a risk committee without any knowledge of the underlying neural architecture. The signs and relative magnitudes of the entries are consistent with established on-chain analysis theory, providing an economic basis for model governance documentation.

Conclusion

This article has presented RS-GARCH-LSTM, an extension of the zhao2024 hybrid GARCH-LSTM framework that replaces a constant scalar mixing weight with a dynamic gate w_t conditioned on six Bitcoin on-chain features. The modification is architecturally minimal, adding seven parameters to the base model, yet produces consistent improvements across all three experimental settings.

On BTC-USD in normal market conditions, RS-GARCH-LSTM is the only model that passes the Kupiec unconditional coverage test at 99% ($p = 0.063$), with a mean VaR 39.8% lower than GJR-GARCH. Under the 2022 bear market scenario, trained exclusively on data preceding the LUNA and FTX events, it achieves the highest Kupiec p-value of any model (0.501), reducing

peak-to-trough drawdown by 33.7 percentage points relative to buy-and-hold. On ETH-USD in a low-volatility test period, it achieves the minimum capital efficiency ratio (4.78× versus 7.56× for GJR-GARCH), a 34.8% reduction in immobilised capital with identical empirical tail protection, and 100% walk-forward MAE consistency across four structurally heterogeneous folds.

The gating weights recovered by end-to-end optimisation exhibit direct economic interpretability: the negative HRATE coefficient on BTC encodes the miner-selling mechanism; the dominant positive NTRBL coefficient on ETH encodes DeFi-specific block saturation; and the sign reversal of HRATE between the two assets reflects the documented BTC-to-ETH capital rotation dynamic. None of these relationships were imposed a priori.

The combination of correct statistical calibration, reduced capital requirements, interpretable model parameters, and real-time data availability makes RS-GARCH-LSTM a deployable tool for institutional cryptocurrency risk management.

References

[1] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. "PyTorch: An Imperative Style, High-Performance Deep Learning Library". In: *Advances in Neural Information Processing Systems 32*. Curran Associates, Inc., 2019, pp. 8024–8035.

[2] Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. "Modeling and Forecasting Realized Volatility". In: *Econometrica* 71.2 (2003), pp. 579–625. DOI: 10.1111/1468-0262.00418.

[3] Andrew J. Patton. "Volatility Forecast Comparison Using Imperfect Volatility Proxies". In: *Journal of Econometrics* 160.1 (2011), pp. 246–256. DOI: 10.1016/j.jeconom.2010.03.034.

[4] William H. Kruskal and W. Allen Wallis. "Use of Ranks in One-Criterion Variance Analysis". In: *Journal of the American Statistical Association* 47.260 (1952), pp. 583–621. DOI: 10.1080/01621459.1952.10483441.

[5] Paul H. Kupiec. "Techniques for Verifying the Accuracy of Risk Measurement Models". In: *Journal of Derivatives* 3.2 (1995), pp. 73–84. DOI: 10.3905/jod.1995.407942.

[6] Peter F. Christoffersen. "Evaluating Interval Forecasts". In: *International Economic Review* 39.4 (1998), pp. 841–862. DOI: 10.2307/2527341.

[7] Alan Moreira and Tyler Muir. "Volatility-Managed Portfolios". In: *Journal of Finance* 72.4 (2017), pp. 1611–1644. DOI: 10.1111/jofi.12513.